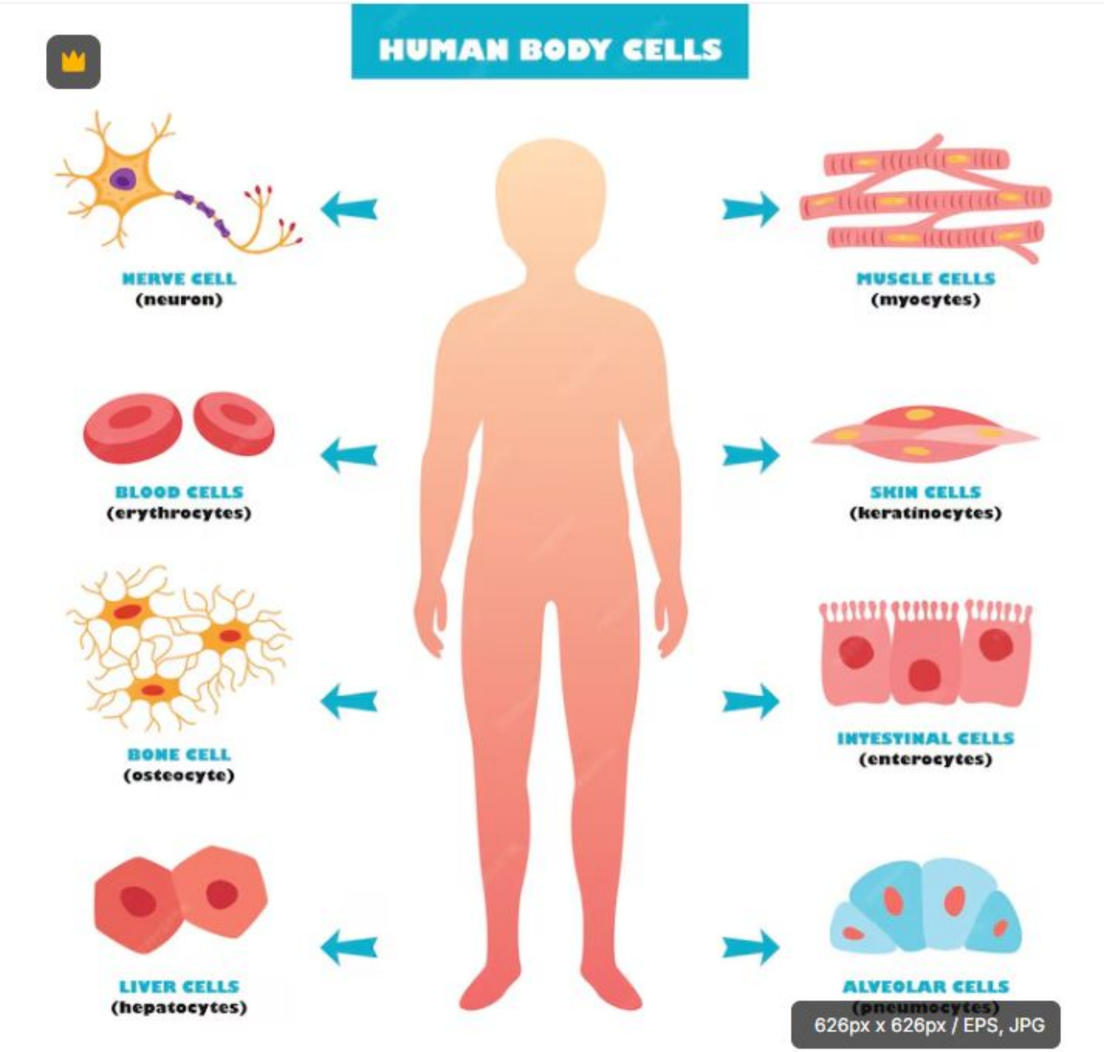
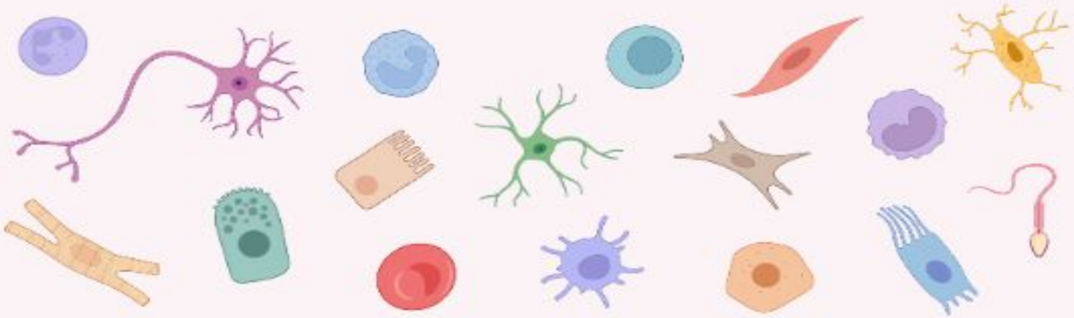
BIOPYTHON

6th sem ISE


HUMAN BODY CELLS
NERVE CELL
(neuron)
MUSCLE CELLS
(myocytes)
BLOOD CELLS
(erythrocytes)
SKIN CELLS
(keratinocytes)
BONE CELL
(osteocyte)
INTESTINAL CELLS
(enterocytes)
LIVER CELLS
(hepatocytes)
ALVEOLAR CELLS
(pneumocytes)
626px x 626px / EPS, JPG

The Different Types of Human Cells

On average, there are **over 30 trillion cells** in a single person! It is estimated that there are **>200 types of cells** with different morphologies and functions.



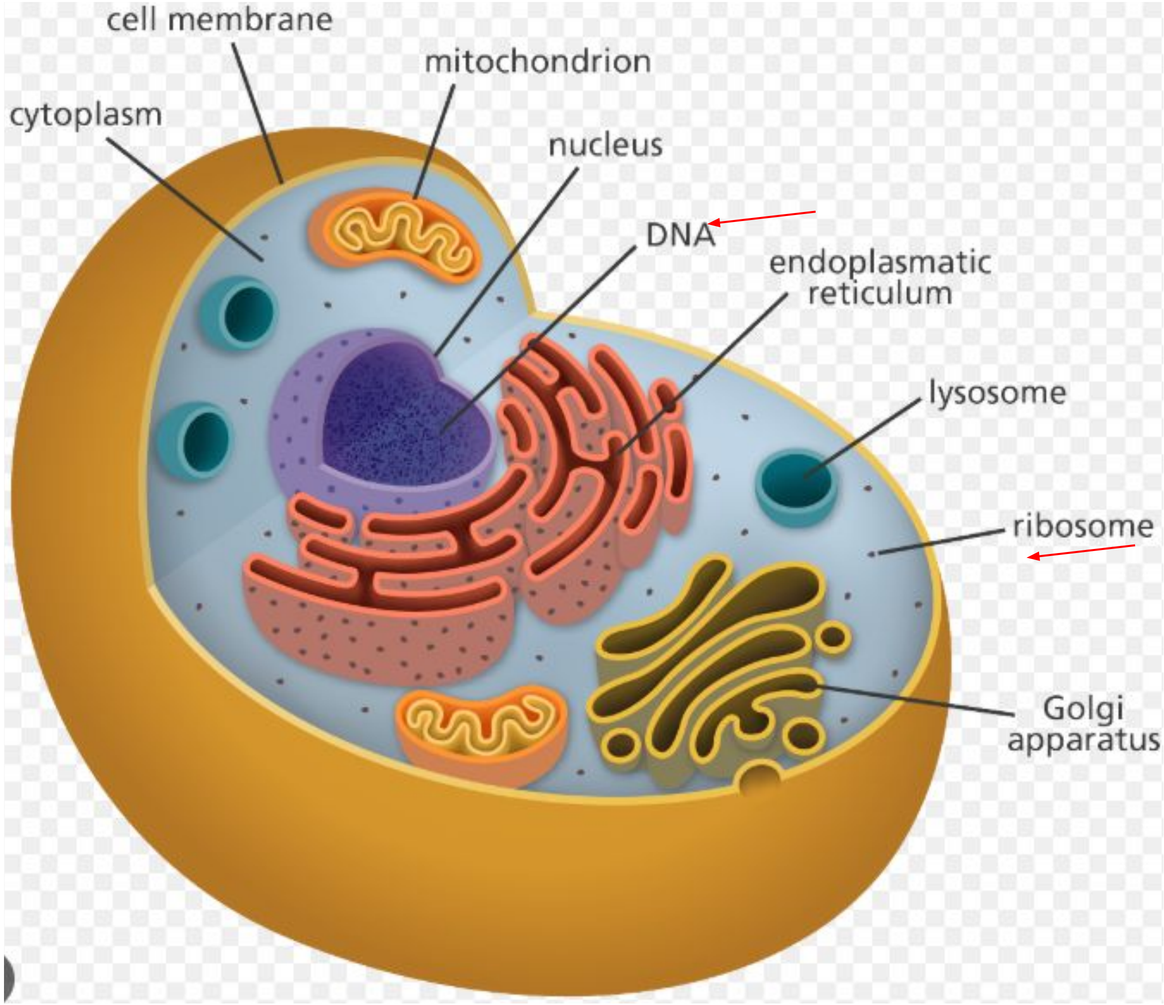
Red blood cell is the most abundant cell type in the human body, accounting for >80% of all cells.

The average lifespan of each type of cell varies greatly. For example:

White blood cells live for ~13 days	**Red blood cells** live for ~120 days	**Neurons** can live as long as the human

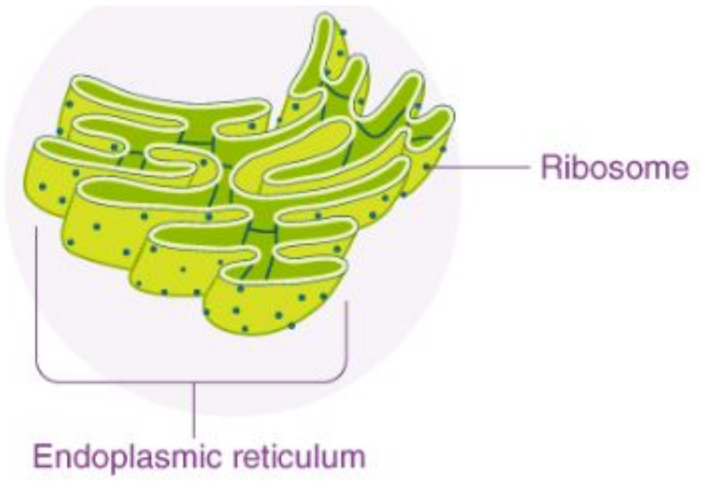
Discover more cell types in the BioRender icon library!

CELL

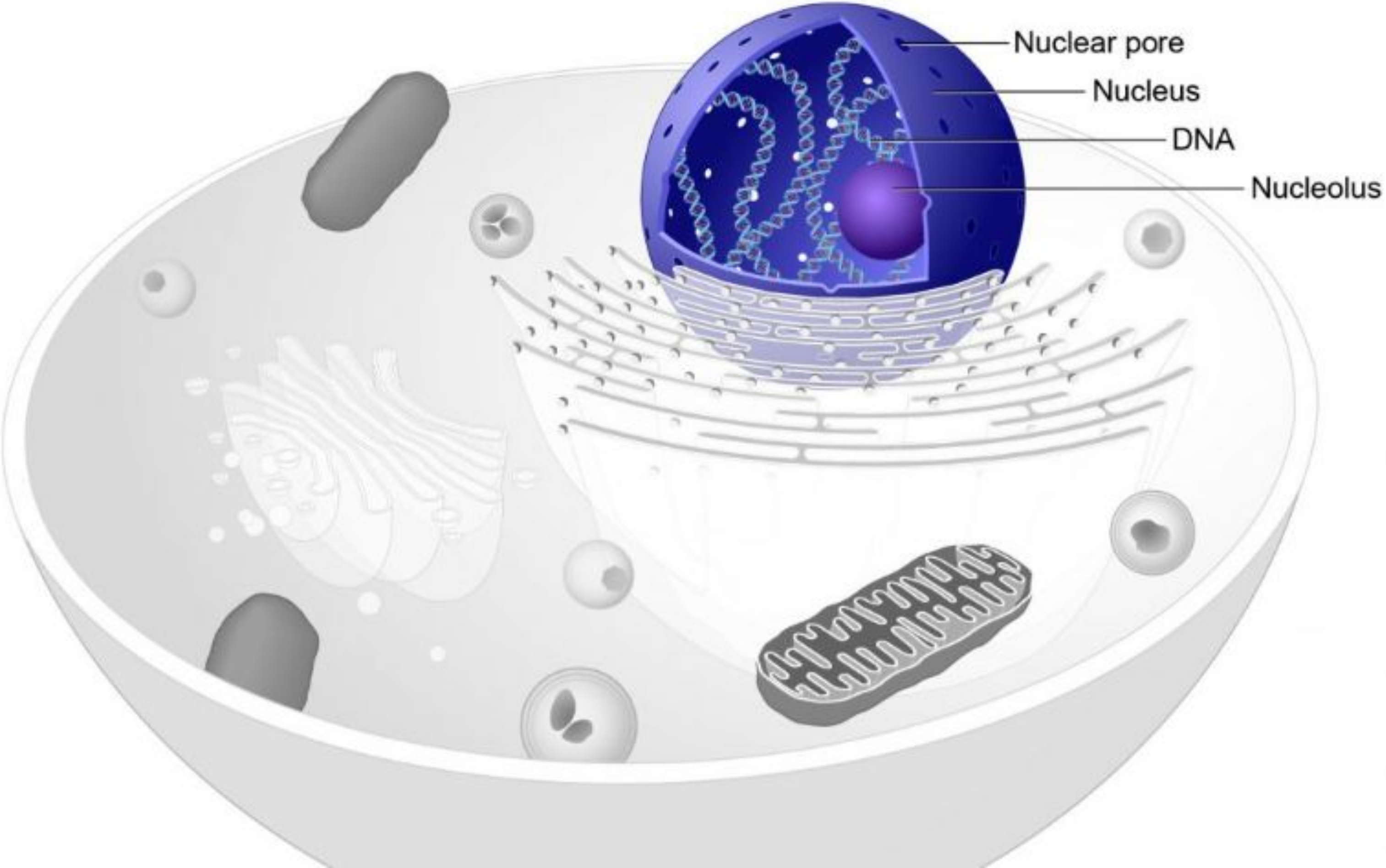

cell membrane
mitochondrion
cytoplasm
nucleus
DNA
endoplasmatic
reticulum
lysosome
ribosome
Golgi
apparatus

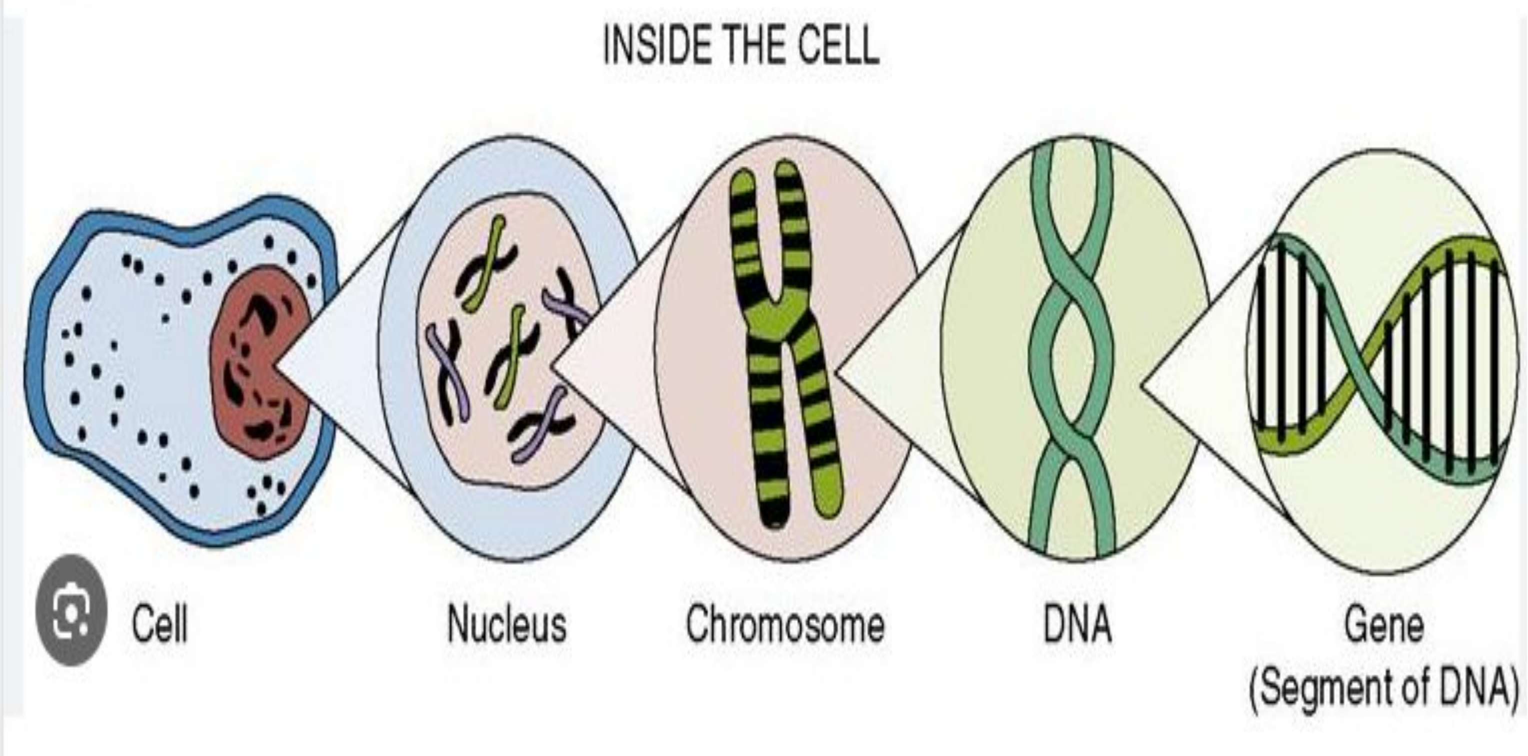
What are Ribosomes?

A ribosome is a complex molecular machine found inside the living cells that produce proteins from amino acids during a process called protein synthesis or translation. The process of protein synthesis is a primary function, which is performed by all living cells.



DNA




INSIDE THE CELL
Cell
Nucleus
Chromosome
DNA
Gene
(Segment of DNA)

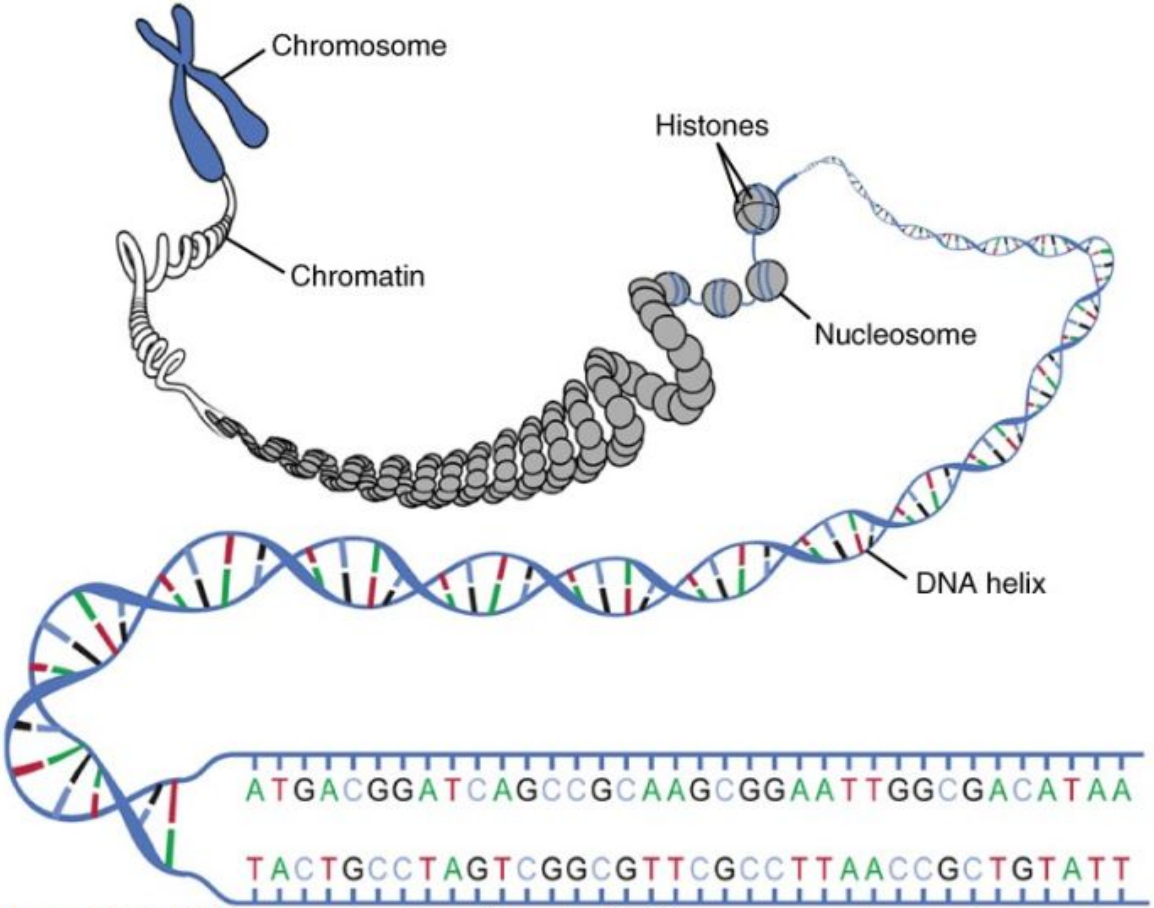
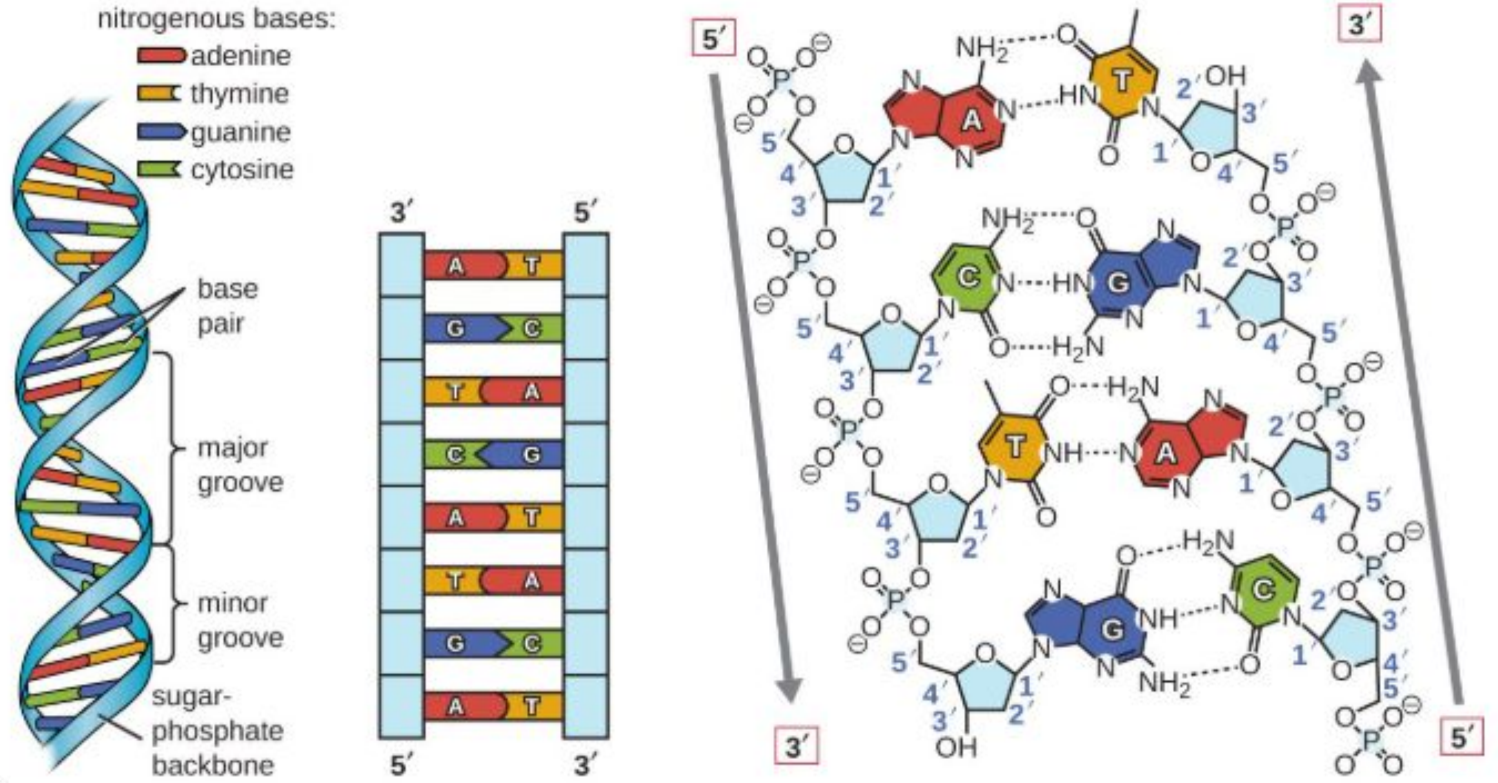
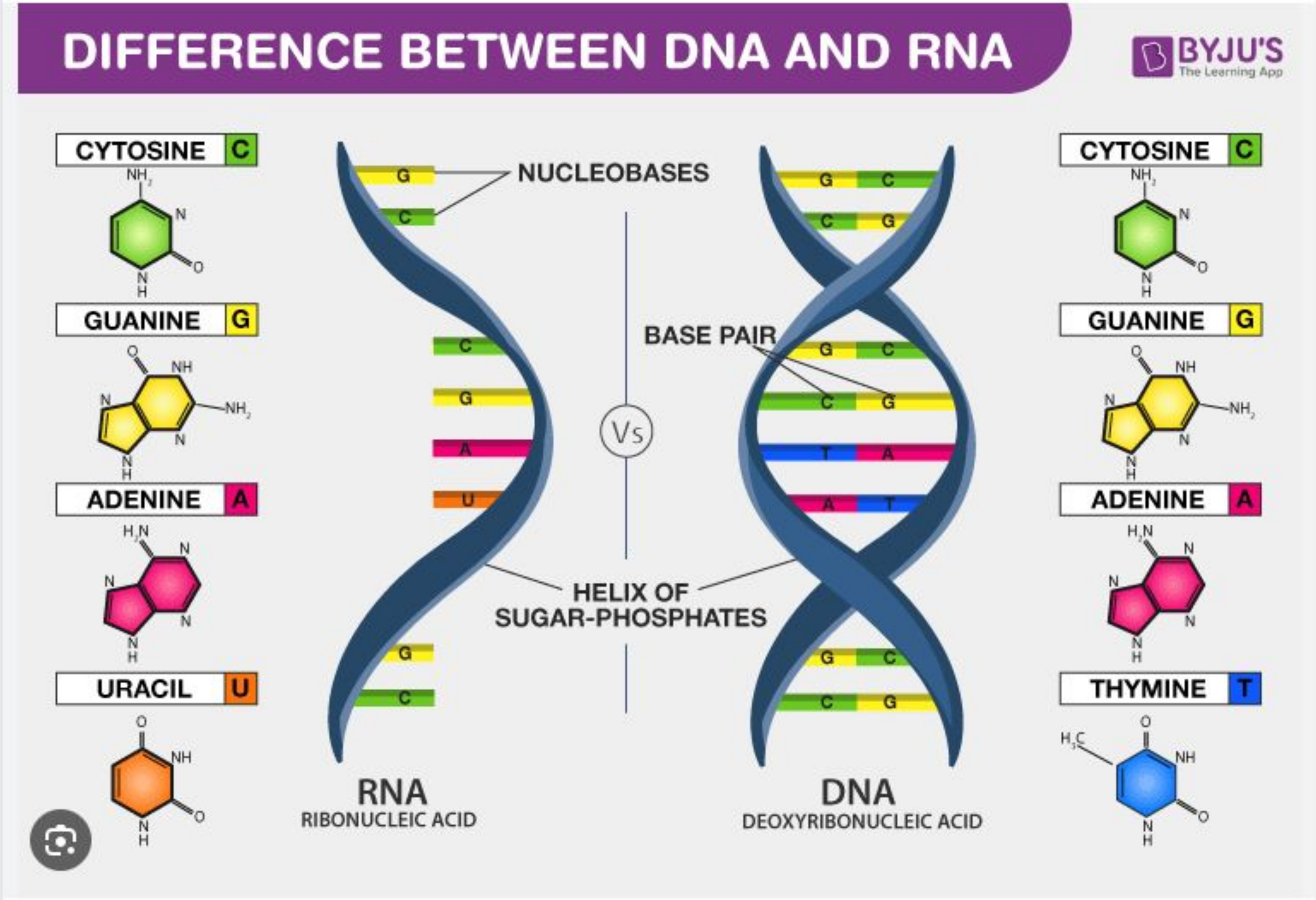
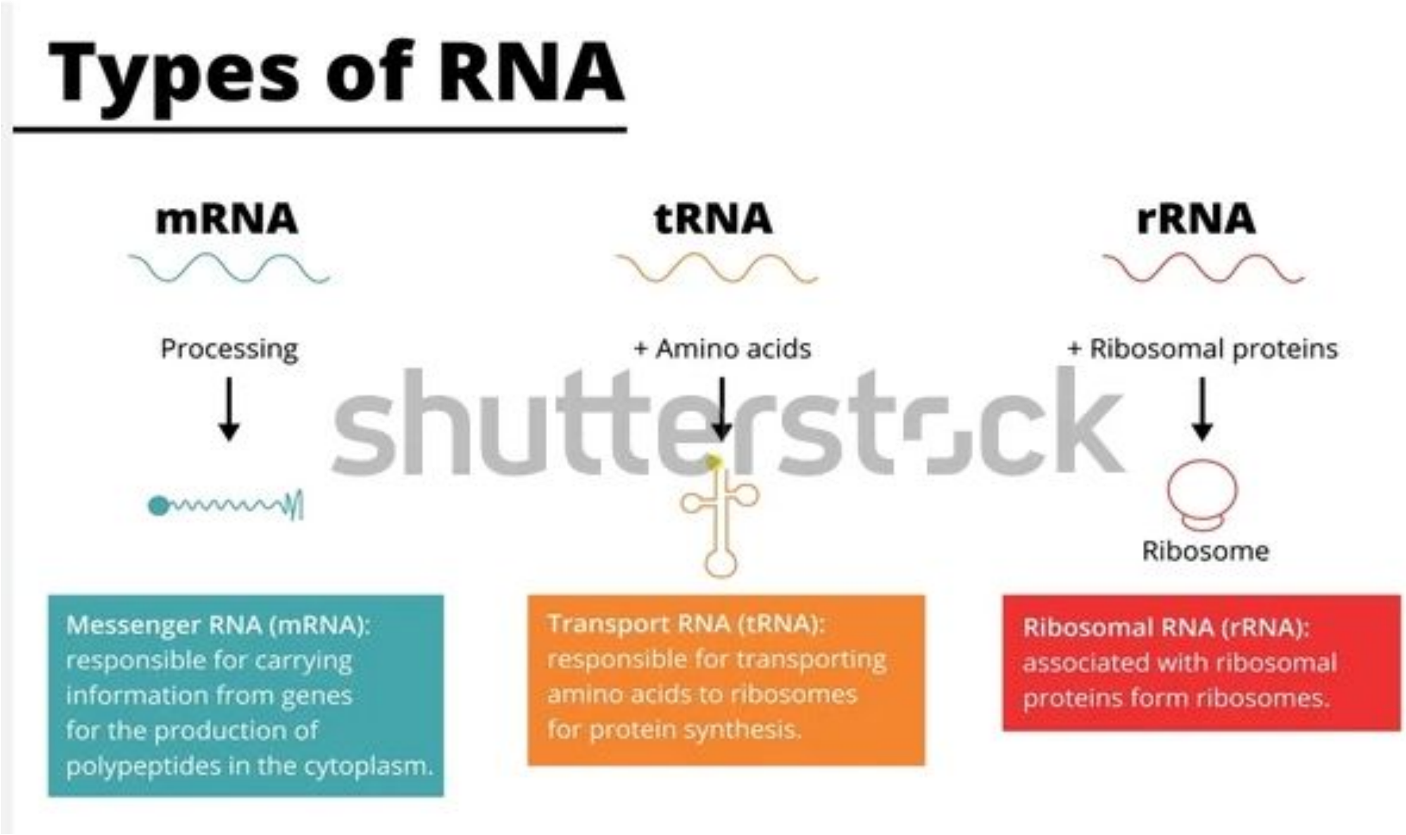


Figure 2.8.4. DNA macrostructure. *Strands of DNA are wrapped around supporting histones. These proteins are increasingly bundled and condensed into chromatin, which is packed tightly into chromosomes when the cell is ready to divide.*



nitrogenous bases:
adenine
thymine
guanine
cytosine
base pair
major groove
minor groove
sugar-phosphate backbone
3′
5′
5′
3′


DIFFERENCE BETWEEN DNA AND RNA
BYJU'S
The Learning App
CYTOSINE C
GUANINE G
ADENINE A
URACIL U
THYMINE T
NUCLEOBASES
BASE PAIR
Vs
HELIX OF
SUGAR-PHOSPHATES
RNA
RIBONUCLEIC ACID
DNA
DEOXYRIBONUCLEIC ACID


Types of RNA
mRNA
tRNA
rRNA
Processing
+ Amino acids
+ Ribosomal proteins
Ribosome
Messenger RNA (mRNA): responsible for carrying information from genes for the production of polypeptides in the cytoplasm.
Transport RNA (tRNA): responsible for transporting amino acids to ribosomes for protein synthesis.
Ribosomal RNA (rRNA): associated with ribosomal proteins form ribosomes.

DNA

- long
- **double**
- Deoxiribose sugar in backbone
- Adenine, thymine, cytosine, guanine
- One kind
- Nucleus
- Hold all the instructions for making proteins

RNA

- **short**
- **single**
- Ribose sugar in backbone
- Adenine, **uracil,** cytosine, guanine
- Three kinds: mRNA, tRNA, rRNA
- Nucleus, and cytoplasm
- Copy the DNA, carry instructions to cytoplasm,
- Line up the Amino acids and bond them together

Macronutrients (Required in large amounts)

Carbohydrates, Proteins, Fats, Water, Fiber

Micronutrients (Required in small amounts)

Vitamins:

Vitamin A, Vitamin B-complex (Vitamin B1 Thiamine, Vitamin B2 Riboflavin, Vitamin B3 Niacin, Vitamin B5 Pantothenic Acid, Vitamin B6 Pyridoxine, Vitamin B7 Biotin, Vitamin B9 Folate/Folic Acid, Vitamin B12 Cobalamin), Vitamin C, Vitamin D, Vitamin E, Vitamin K

Minerals:

Calcium, Iron, Magnesium, Zinc, Phosphorus, Potassium, Sodium, Copper, Iodine, Selenium, Manganese, Fluoride

PROTEIN

Proteins are **long chains of amino acids** linked together by **peptide bonds**. The sequence and arrangement of amino acids determine the structure and function of a protein.

Amino Acids: Building Blocks of Proteins

Amino acids are organic molecules that serve as the **building blocks of proteins**. Each amino acid consists of:

- **A central carbon (C)**
- **An amino group ($-NH_2$)**
- **A carboxyl group (-COOH)**
- **A hydrogen atom (H)**
- **A unique side chain (R-group)** that determines the amino acid's properties

There are **20 standard amino acids** in the human body, classified into **essential** (must be obtained from food) and **non-essential** (produced by the body).

Essential Amino Acids (Must be obtained from food)

Essential Amino Acid	Food Sources
Histidine (His)	Meat, fish, poultry, dairy, soybeans
Isoleucine (Ile)	Eggs, fish, nuts, seeds, lentils
Leucine (Leu)	Beef, chicken, dairy, peanuts, soybeans
Lysine (Lys)	Meat, fish, eggs, dairy, quinoa, legumes
Methionine (Met)	Eggs, fish, chicken, Brazil nuts, sesame seeds
Phenylalanine (Phe)	Dairy, meat, poultry, soy, nuts
Threonine (Thr)	Cottage cheese, lentils, turkey, wheat germ
Tryptophan (Trp)	Turkey, dairy, chocolate, oats, nuts
Valine (Val)	Meat, dairy, soy, peanuts, mushrooms

Non-Essential Amino Acids (Body can synthesize but also found in food)

Non-Essential Amino Acid	Food Sources
Alanine (Ala)	Meat, eggs, dairy, soy, whole grains
Arginine (Arg)	Nuts, seeds, meat, fish, dairy
Asparagine (Asn)	Dairy, eggs, fish, nuts, legumes
Aspartic Acid (Asp)	Sugarcane, poultry, fish, beans
Cysteine (Cys)	Eggs, garlic, onions, poultry, broccoli
Glutamic Acid (Glu)	Tomatoes, cheese, soy sauce, meat
Glutamine (Gln)	Beef, chicken, dairy, spinach, cabbage
Glycine (Gly)	Gelatin, poultry, fish, dairy, legumes
Proline (Pro)	Meat, dairy, eggs, collagen-rich foods
Serine (Ser)	Soybeans, nuts, eggs, dairy, legumes
Tyrosine (Tyr)	Cheese, turkey, fish, nuts, soy products

EXAMPLE:

The **amino acid** that plays a crucial role in **blood clotting** is **Glutamic Acid (Glu)**. It is involved in the activation of clotting factors, particularly through its modified form **γ-carboxyglutamate (Gla)** in proteins like prothrombin.

The **protein** primarily responsible for blood clotting is **Fibrinogen**, which is converted into **Fibrin** by the enzyme **Thrombin** to form a clot.

Protein from Transcription & Translation (Internal Production)

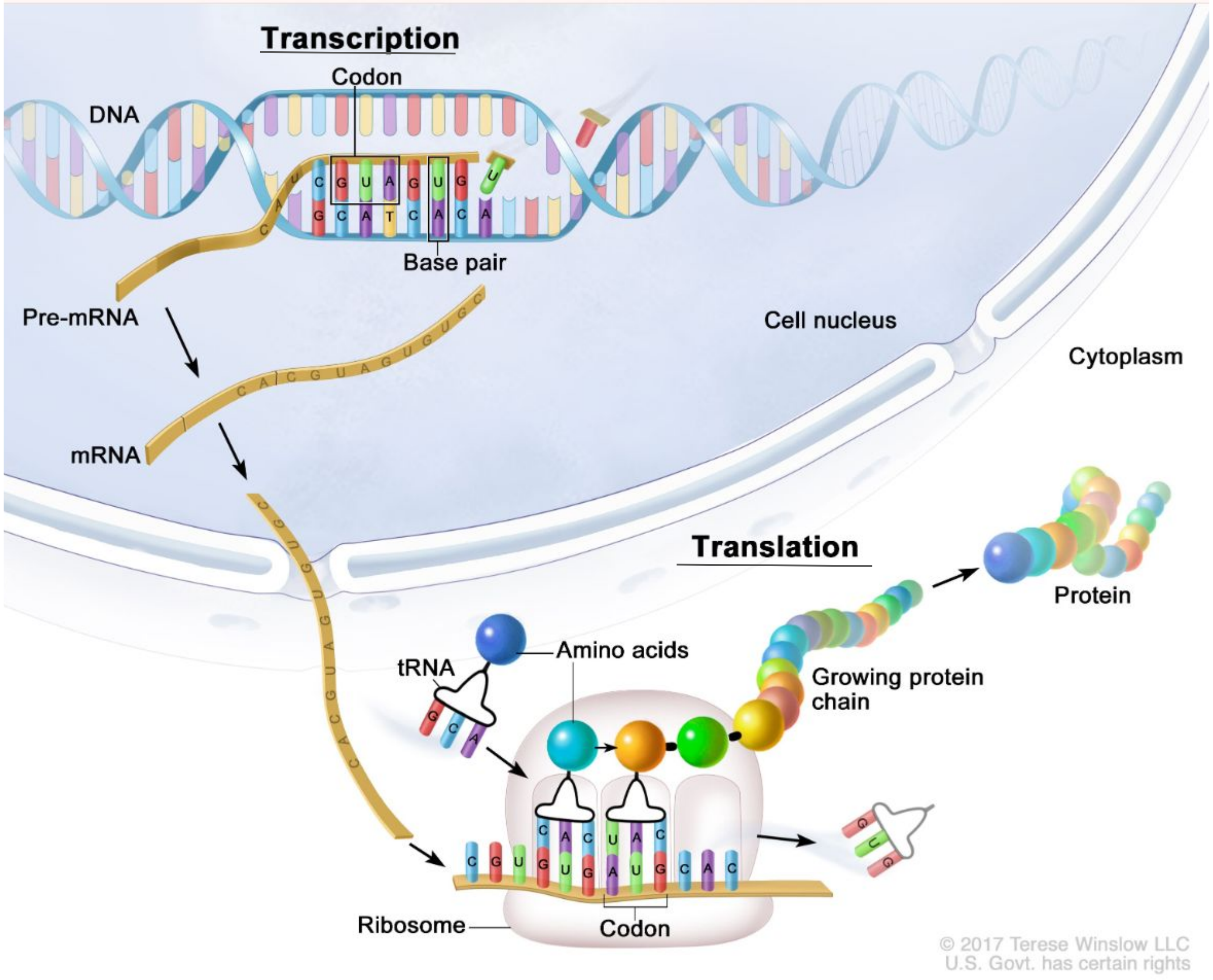
Even though we consume protein from food, our body needs to **make its own proteins** for specific functions like enzymes, hormones, and structural proteins. This happens through **transcription and translation**:

Step 1: Transcription (DNA → mRNA)

- Inside the cell nucleus, DNA holds the instructions for making proteins.
- A specific **gene** is copied into **messenger RNA (mRNA)**.
- The mRNA carries this genetic information to the ribosomes in the cytoplasm.

Step 2: Translation (mRNA → Protein)

- Ribosomes read the mRNA sequence and assemble amino acids into a **polypeptide chain**.
- **Transfer RNA (tRNA)** brings the correct amino acids.
- The chain folds into a **functional protein** used by the body.


Transcription
Codon
DNA
Base pair
Pre-mRNA
Cell nucleus
Cytoplasm
mRNA
Translation
Protein
tRNA
Amino acids
Growing protein chain
Ribosome
Codon
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Protein synthesis /DNA transcription & translation

https://www.youtube.com/watch?v=bKIpDtJdK8Q

Here are all the different codon combinations that encode each of the 20 standard amino acids:

1. Alanine (A)

- GCU
- GCC
- GCA
- GCG

2. Arginine (R)

- CGU
- CGC
- CGA
- CGG
- AGA
- AGG

3. Asparagine (N)

- AAU
- AAC

4. Aspartic acid (D)

- GAU
- GAC

5. Cysteine (C)

- UGU
- UGC

6. Glutamic acid (E)

- GAA
- GAG